26 October 2026

EEG AI Honours Project Kickoff

A project for one student, with room for a team

Project kickoff

01	A clear question and a useful demo	Student
02	Existing work we can reuse	Student
03	Monash resources and dataset access	Supervisor
04	One controlled model comparison	Student
05	Milestones and optional team roles	Team
06	Learning, risks and first decisions	Together

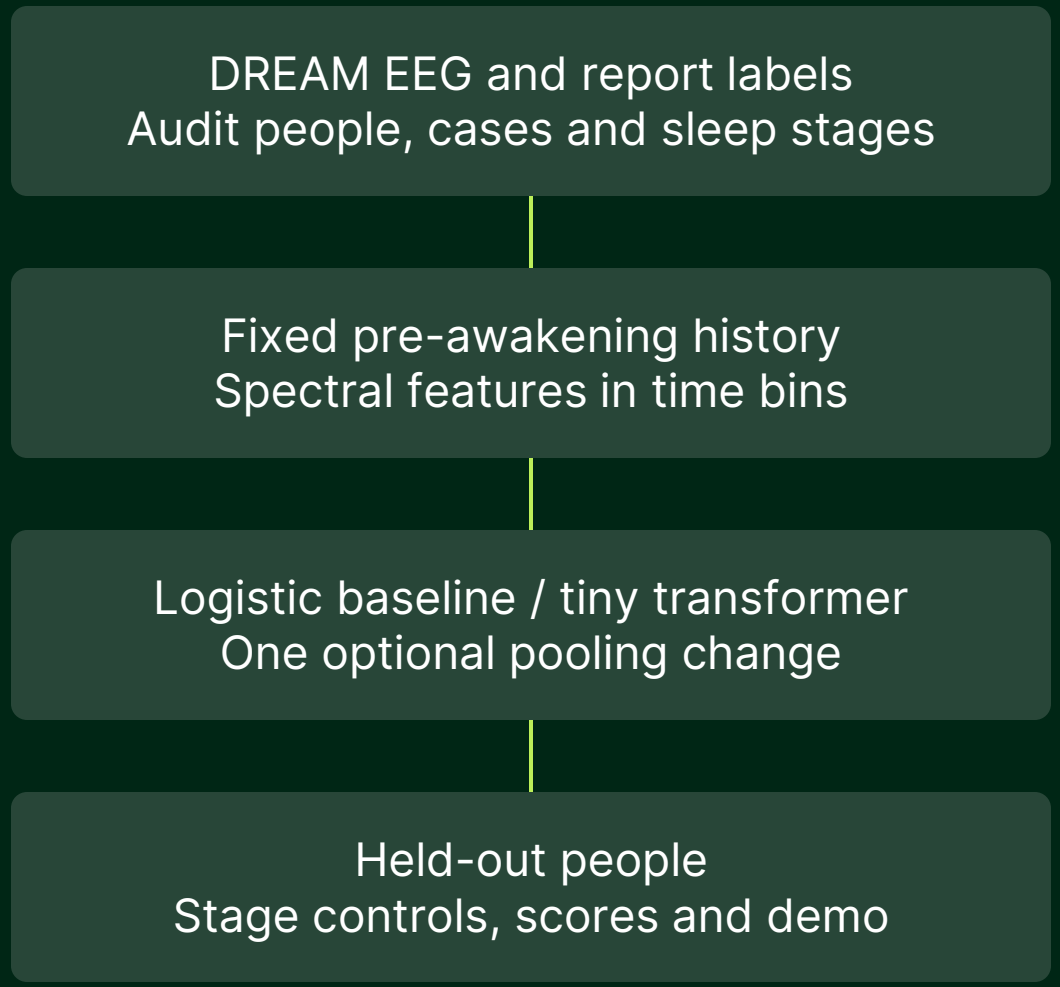
The core project

A question we can answer

Can EEG features predict reported sleep experience in unseen participants? Compare a simple spectral classifier with a tiny transformer.

A result we can demonstrate

Build an EEG model-comparison viewer. Show the features, report labels, predictions and failure cases. Keep EEG-to-BOLD as a separate extension.



One question, three concrete deliverables

BASELINE

A reproducible baseline

Audit one EEG subset. Fit a spectral logistic classifier with participant-grouped splits. Include majority and sleep-stage controls.

MODEL CHANGE

One transformer study

After the baseline works, compare mean pooling with learned time-bin weighting. Keep the same features and split. A clear negative result counts.

DEMO

An explanatory demo

Show signals, scores and failures. Include a short report and a walkthrough the student can give without relying on AI-written explanations.

Monash resources worth using

M3 and eResearch

Reuse the working cluster connection. Confirm CI-led project access and limits. Use CPU baselines, a GPU pilot and HPC training or drop-ins.

Biomedical Imaging

MBI lists EEG and simultaneous EEG-fMRI support. Treat new recordings as optional, with facility training, ethics, bookings and a budget.

DREAM and Turner

Start with a public DREAM subset. Seek a sleep/EEG adviser through a supervisor. A Monash-hosted collection is not automatically Monash-collected data.

IT and DeepNeuron

The IT EEG representation-learning project is a supervision lead. DeepNeuron offers a possible AI/HPC student community. Places and collaboration need confirmation.

One public subset, with a workable fallback

DREAM is the starter
Paired fMRI is an extension

Zhang and Wamsley subset: 28 participants and 308 catalogued cases, under CC BY 4.0. The collection is Monash-associated; this cohort was collected at Furman University.

Audit usable cases and sleep-stage confounding before training.

Fallback: Sleep-EDF

Use one EEG derivation and scored sleep stages. Keep both nights from each person in the same partition.

Extension: ds006040

Reuse the paired EEG-fMRI pipeline if needed. It released 28 people; our earlier analysis retained 27.

One student first, a team when useful

One student owns the complete small pipeline. A team of three or four can divide these roles once the baseline works. Each person should explain the shared experiment and identify their own contribution.

Data and QC	A dataset card, fixed EEG windows, participant splits and a small quality-control report.
Models	A spectral classifier and tiny transformer sharing the same inputs and prediction format.
Evaluation	Scores with uncertainty over people, sleep-stage controls and a leakage audit.
Demo and guide	A results viewer, failure examples, installation steps and the advisor presentation.

An example 12-week starting phase

Weeks 1-4

A baseline we understand

Confirm labels and splits. Plot the EEG. Fit the spectral classifier and check sleep-stage controls.

Weeks 5-8

One controlled study

Run a GPU pilot. Freeze the comparison. Test pooling and inspect errors before considering an extension.

Weeks 9-12

A result we can communicate

Package the demo and code. Write the report. Present the limitations and a justified next step.

A controlled study with a fixed budget

Proposed starting limits

Begin with CPU baselines and fewer than 20,000 transformer parameters. Include majority and sleep-stage controls. Select on validation; report balanced accuracy, macro F1 and uncertainty over people. Freeze a budget before evaluation. These are teaching choices.

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Logistic classifier, tiny transformer and pooling variant.

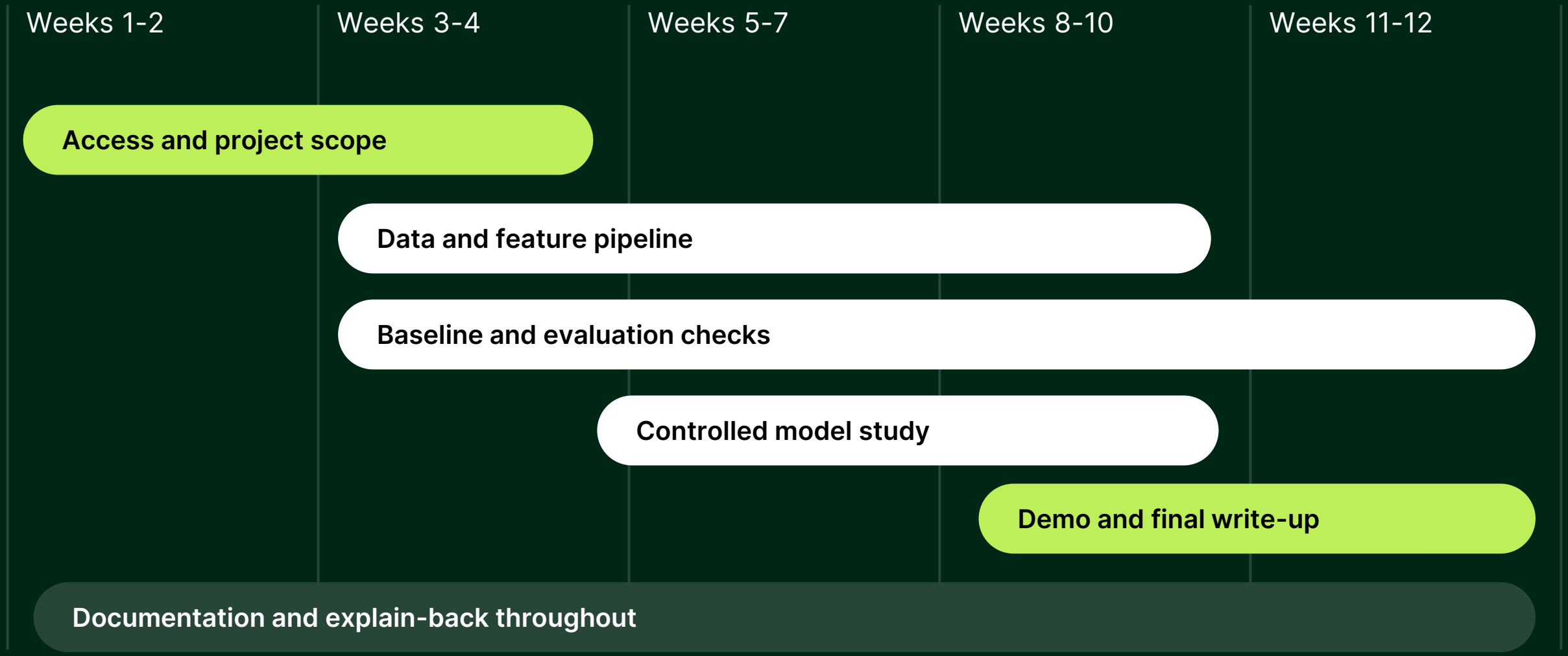
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One pooling change after the baseline and data checks work.

24 h

Proposed GPU-hour ceiling for the initial study. Review after one pilot fit.

Milestones and checkpoints



A four-week learning and review cycle

Monday	Tuesday	Wednesday	Thursday	Friday
Week 1 **Data and permissions** **Explain one window**		**Baseline walkthrough**		
Week 2				
Week 3 **Pooling experiment**			**Failure-case review**	
Week 4	**Supervisor demo**			

Honours project proposal

Start small
Explain the result

Next decision: dataset, supervisor and available time